

Design procedure for two way slabs

Step 1: Assume the depth of the slab based on stiffness

(a) for two way slabs with shorter span less than 3.5m and live load less than 3.5 kN/m the allowable l_x/d ratio

	Fe 250	Fe 415
simply supported beam	35	28
continuous cantilever beam	40	32

b) if l_x greater than 3.5m and live load greater than 3.5 kN/m the allowable l_x/d ratio is same as that of in one way slab (Pg) 35; 40 l_x = shorter span
 l_y = longer span

Step 2: find the effective spans l_x and l_y

Step 3: calculate the ultimate load considering 1m width of the slab

Step 4: obtain the design moment coefficients along short and long span depending upon boundary conditions which is given in table No: 26 of IS 456.
calculate the bending moment

step 5: calculate the minimum depth required to resist the maximum design moment which should be less than the depth provided otherwise increase the depth.

step 6: calculate the area of the steel at the mid span (and at support if the slab is continuous) in both the directions. (pg: 96)

The shorter span bars are provided in the bottom layer and along span bars are provided above the short span bars in the mid span region. Therefore for short span

$$d = D - \text{clear span} - \left(\frac{\text{diameter}}{2}\right)$$

for long span

$$d_1 = \left(D - \left(\text{clear span} - \frac{\text{diameter}}{2}\right) - \text{dia}\right)$$

$$d_1 = d - \text{diameter}$$

Step 4 The main reinforcement shall be provided in the middle strip of width equal to $\frac{3}{4}$ of slab width.

step 7: Torsion steel.

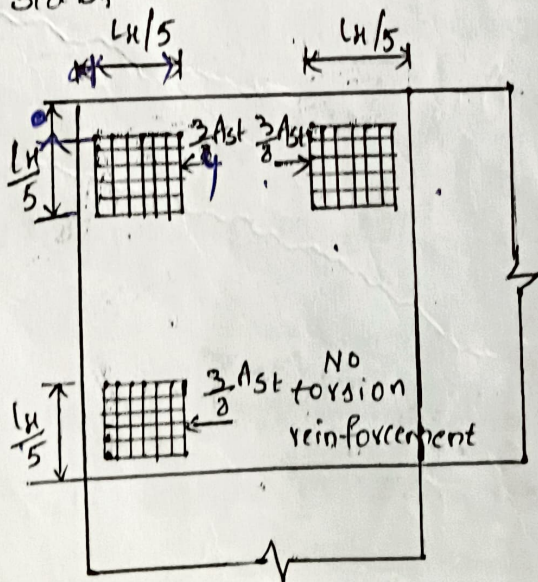
(a) At corners where slab is discontinuous over both edges $A_t = \frac{3}{4} A_{stx}$

b) At corners where slab is discontinuous over one edge. $A_t = \frac{3}{8} A_{stx}$

c) At corners where slab is continuous over both the edges $A_t = 0$ i.e. no torsion steel is required

Where A_{stx} = Area of steel for maximum mid span moment

d) The area of torsion reinforcement will be provided at the corners in the form of mesh, one at top and other at bottom for the length of $L_x/5$ in each orthogonal direction parallel to the sides of the slabs



Step 8: check for deflection

Step 9: check for shear

Step 10: check for development length

Note: The check for shear and check for development length are mostly satisfied in all cases of slabs subjected to the uniformly distributed loads and therefore we neglect these checks in design calculations

1. Design a two way slabs for room 4000mm X 3500mm in size, if the super imposed load is 3 kN/m^2 and floor finish of 1 kN/m^2 the edges of the slab are simply supported and corners are not held down. Use M20 grade and FE415.

Solⁿ $\frac{L_y}{L_x} = \frac{4000}{3500} \Rightarrow 1.14$

$\frac{L_y}{L_x}$ ratio is less than 2.

we have to design for two way slabs

Step 1: $L_x = 3500 \text{ mm}$

$L_y = 4000 \text{ mm}$

live load = 3 kN/m^2

floor finish = 1 kN/m^2